

AI Device Inspector Platform.

Photograph a laptop, let the model find the damage, and get a defensible grade in under a minute — with every decision recorded.

PREPARED FOR

Hugo Martinez

Project sponsor

PREPARED BY

Pixorama Group

Product engineering

12

WEEKS TOTAL

8

WEEKS BUILD

4

WEEKS TEST

12

MODULES

60

FEATURES

Thermal Failure · 99%

Bent Chassis · 91%

Chipped Casing · 84%

F

GRADE · RECYCLE

What this platform actually does

A used laptop goes in. A trusted grade, a photo record and a price signal come out.

Cosmetic & functional grade	A-F, from one rule set
Health score	0-100, weighted by defect
Photographic evidence	Six angles, defects marked
Recommended disposition	Resell · repair · parts · recycle
Audit trail	Every decision, attributable

Refurbishment centres take in thousands of second-hand devices. Someone has to decide what condition each one is in — is it a clean resale, does it need a repair, or is it only worth parts? That decision sets the price.

Today that job is done by eye. An operator turns the laptop over, notes the scratches on a form, guesses a grade, and moves on. Two people looking at the same device often disagree. When a buyer complains, there is no evidence to point at — just a line in a spreadsheet.

This platform replaces the eye with a camera booth and a trained model. The operator scans the serial, the device is photographed from six angles, and the model marks every defect it finds — scratches, dents, cracked screens, missing keys, swollen batteries — with a confidence score and a severity. The operator confirms or corrects what they see, and the system produces a grade, a health score, and a printable report with the photos attached.

The result is that grading stops depending on who is working that shift. It becomes consistent, fast, and — because every click is logged — something you can defend to a buyer, an auditor, or an insurer.

● HOW IT WORKS TODAY

- ❑ Grading is a judgement call – the same laptop gets a B from one operator and a C from another.
- ❑ Condition notes live in a spreadsheet with no photographs attached to them.
- ❑ A buyer dispute takes hours to investigate and usually ends in a credit note.
- ❑ Nobody can tell you which supplier or model is sending in the worst stock.
- ❑ Training a new operator to grade reliably takes weeks.

● HOW IT WORKS ON THE PLATFORM

- ❑ One model applies one rule set to every device, on every shift, in every facility.
- ❑ Six photographs and a marked-up defect map are attached to every record, permanently.
- ❑ A dispute is answered by sending a link to the inspection – photos, defects, grade, and who signed it off.
- ❑ Defect trends by model, supplier and intake batch are visible from day one.
- ❑ A new operator follows a guided six-step flow and is productive on their first day.

60s

Capture to grade, per device

TARGET

6

Camera angles per inspection

FIXED RIG

90%

Detection recall on top defect classes

ACCEPTANCE GATE

100%

Grade changes carrying an audit entry

NON-NEGOTIABLE

02

THE USE CASE

One device, end to end

This is the journey the whole product is built around. Everything else in the feature list exists to make one of these seven steps faster, more accurate, or provable after the fact.

01

Intake & identification

A pallet arrives. The operator scans the barcode or lets the camera read the service tag. The platform pulls the model, original spec and any prior inspection history for that serial, so a device that has been round the loop before is flagged immediately.

OPERATOR • 5 SECONDS

02

Guided positioning

The device is placed in the booth. On-screen guides show the operator exactly where it goes, lid open or closed, for the model in question. The station checks lighting and focus before it will let the capture start.

OPERATOR • 10 SECONDS

03

Six-angle capture

One trigger fires the whole array – top, bottom, left, right, front and keyboard. Blurred or over-exposed frames are rejected on the spot and re-shot, so a bad photo never becomes a bad grade.

AUTOMATED • 8 SECONDS

04

AI analysis

Images go to the inference service. The model segments each defect, classifies it against the taxonomy, scores its confidence and assigns a severity. A heat map and a set of bounding regions come back attached to the record.

AUTOMATED • 12 SECONDS

05

Operator review

The operator sees the marked-up photographs. They accept what is right, reject false positives, merge duplicates, resize a region that is slightly off, or draw a new one for something the model missed. Every one of those actions is written to the audit trail with a before and after value.

OPERATOR • 20 SECONDS

06

Grade & disposition

The grading engine turns the confirmed defect set into a cosmetic grade, a functional grade and a health score, using the rule profile for that buyer. It recommends resell, repair, parts or recycle. A supervisor can override, but only with a reason attached.

AUTOMATED + SUPERVISOR • 5 SECONDS

07

Report, hand-off & feedback

A branded PDF certificate is generated and the record is pushed to the ERP by webhook. The buyer gets a read-only link. Meanwhile every operator correction is added to the training set, so next month's model is measurably better than this month's.

AUTOMATED • INSTANT



03

WHO USES IT

Four people, four different products

The same data, surfaced very differently depending on who is looking. Permissions are enforced in the API, not just hidden in the interface.

ROLE 01

Operator

Lives in the station screen. Scans, captures, reviews detections. Needs speed, big hit targets and a flow that works with gloves on and a queue behind them.

ROLE 02

Supervisor

Watches the floor in real time, picks up low-confidence and disputed inspections, approves grade overrides, and answers buyer queries with evidence.

ROLE 03

Administrator

Configures clients, stations, users, defect taxonomy and grading profiles. Owns the integrations, retention policy and the audit export.

Client / buyer

Read-only. Sees their own consignments, the grades and the photographic evidence. Never sees another client's data — isolation is enforced at the database row.

04

SCOPE

The feature list

Twelve modules, sixty features. This is the full v1 scope — everything below is committed to the eight-week build and verified in the four-week test window. Anything not on this list is a change request, not an assumption.

01 Access, roles & tenancy

Who can see what, across multiple client accounts on one deployment.

- Email & password sign-in with optional two-factor
- Enterprise SSO via SAML 2.0 / OIDC (Entra ID, Okta)
- Four roles — Operator, Supervisor, Admin, Client — with a published permission matrix
- Multi-tenant isolation enforced with Postgres row-level security
- Station kiosk sessions: device-bound tokens, idle lock, fast operator switch

02 Dashboard & live floor

One screen that tells a supervisor whether today is going well.

- Live station tiles — idle, capturing, analysing, needs review, fault
- Shift KPIs: units processed, average cycle time, first-pass yield, rework rate
- Throughput trend and defect-mix charts, click-through to a filtered list
- Alert feed for offline stations, calibration due and low-confidence runs
- Operator activity summary for the current shift

03 Inspection station

The capture workflow the operator actually spends their day in.

- Six-state guided flow: scan → verify → position → capture → analyse → review
 - Barcode / QR scan plus OCR service-tag reading, with manual entry fallback
 - Automatic device lookup: model, spec, prior inspections, warranty status
 - Six-camera trigger with per-angle retake, focus and exposure validation
 - Offline-tolerant: captures queue locally and sync when the network returns
-

04 AI detection engine

The model layer — what it looks for and how sure it is.

- Instance segmentation and classification across cosmetic, display and hardware defects
 - Taxonomy of 20+ classes: scratch, dent, crack, chip, bent chassis, missing key, dead pixel, backlight bleed, hinge play, port damage, corrosion, thermal failure
 - Per-detection confidence, four-level severity, bounding region and heat map
 - Configurable confidence thresholds with auto-escalation to supervisor review
 - Model versioning — every result is stamped with the exact model that produced it
-

05 AI review & annotation

Human-in-the-loop correction, without slowing the line down.

- Annotated viewer with bounding boxes, heat map toggle and zoom-to-defect
- Accept, reject, reclassify, change severity, merge duplicate detections
- Cursor tool bubble — resize an AI region or draw a missed one directly on the photo
- Low-confidence queue with second-opinion routing to a supervisor
- Every correction written to the audit trail and fed back into the training set

06 Grading & disposition

Turning a confirmed defect set into a commercial decision.

- Separate cosmetic and functional grades (A–F) from a configurable rule set
 - Weighted health score, 0–100, with the contributing defects shown
 - Per-client grading profiles — one buyer's B is another buyer's C
 - Suggested disposition: resell, repair, harvest for parts, or recycle
 - Supervisor override with mandatory reason code, captured in the record
-

07 Inspection history

Finding any device, out of hundreds of thousands, in seconds.

- Sixteen-column data grid, virtualised for 100k+ records
 - Global search plus advanced filters with saved, shareable presets
 - Bulk actions: approve, reject, assign, archive, re-run AI, export
 - Inline editing gated by role, with per-field permission checks
 - Column chooser, density toggle and per-user saved views
-

08 Records & reporting

The evidence pack that settles a dispute.

- Tabbed record: overview, images, AI results, audit trail, edit
- Full-resolution image viewer with drag-to-reorder image sets
- Branded PDF certificate — grade, defects, photographs, operator, model version
- Expiring read-only share link for buyers and insurers
- Tamper-evident content hash printed on every certificate

09 Data in / out

The platform has to live inside your existing systems, not beside them.

- Five-step CSV import wizard: template, mapping, validation, preview, commit
- Export to CSV, XLSX, PDF and JSON, filtered or bulk-selected
- Documented REST API with OpenAPI spec and signed webhooks
- ERP / WMS connector plus label printing (ZPL) from the station
- Scheduled buyer feeds pushed to SFTP or S3 on a cron

10 Analytics

Answering "where are we losing money?" rather than "how many did we do?".

- Throughput, cycle time, first-pass yield and rework, by station and shift
- Grade distribution over time, by model and by intake batch
- Defect Pareto by supplier — which source is sending the worst stock
- AI-versus-human agreement rate and the most common override reasons
- Custom date ranges, CSV export and a scheduled email digest

11 Administration

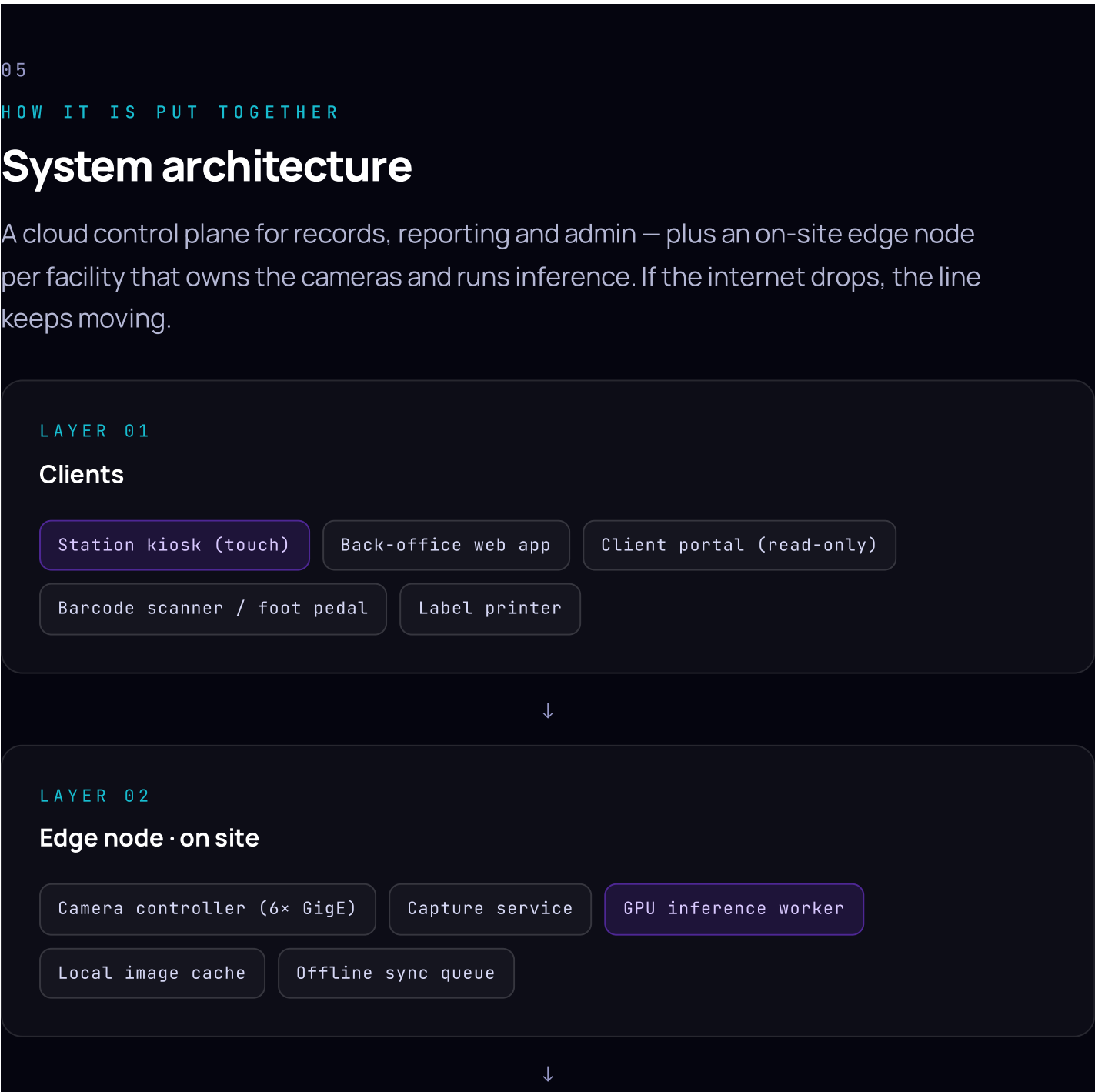
Configuration your team owns, without calling us.

- Client, user, role and station management with camera calibration profiles
- Defect taxonomy and severity-rule editor
- Grading profile builder with a live preview against historic records
- Notification rules and per-client report branding
- Feature flags per tenant for staged rollout

12 Audit & compliance

Being able to prove, months later, exactly what happened.

- Immutable audit log — actor, action, before, after, timestamp — fully exportable
- Per-client retention and purge policy with a right-to-erasure workflow
- Encryption in transit and at rest, with a documented key rotation policy
- Field-level redaction for client-role users
- SOC 2-aligned control set with evidence export for auditors



LAYER 03

Core API

NestJS · TypeScript

REST + OpenAPI

WebSocket gateway

BullMQ job queue

Webhook outbox

PDF render worker



LAYER 04

Data & models

PostgreSQL 16 · RLS

S3 / MinIO object store

Redis

MLflow model registry

Training data lake



LAYER 05

Platform

Docker · Compose / K8s

Terraform

GitHub Actions CI/CD

OpenTelemetry → Grafana

Sentry

Nightly PITR backups

06

BUILD DETAIL

The front end

One codebase serves three very different surfaces: a touch kiosk on the floor, a dense back-office application, and a read-only portal for buyers.

The station screen is dark by default — it sits under industrial lighting and is looked at for eight hours straight. Hit targets are sized for gloved hands, the whole capture flow is reachable from the keyboard or a foot pedal, and the interface stays responsive while six full-resolution images upload in the background.

The back office is the opposite problem: a hundred thousand records, sixteen columns, and a supervisor who needs to find one serial number. That is a virtualised table, server-side filtering and saved views — not an infinite scroll.

DESIGN SYSTEM FIRST

Week one produces a token set — colour, type scale, 8px spacing, radius, motion timing — and the component library built on it. Every screen after that is assembled from those parts, which is what keeps eight weeks of build looking like one product rather than four.

Framework

[Next.js 16](#) · [React 19](#)

Server components keep record pages fast; one app covers kiosk, back office and portal.

Language

[TypeScript \(strict\)](#)

Types shared end to end with the API contract — no drift between client and server.

Styling

[Tailwind v4 + design tokens](#)

Single source of truth for the visual system; dark-first for the floor.

Client state

[Zustand](#)

Small, predictable stores for UI and station state. No boilerplate.

Server state

[TanStack Query](#)

Caching, background refetch and optimistic updates for review actions.

Tables

[TanStack Table \(virtualised\)](#)

100k+ rows at 60fps with column control and server-side sort/filter.

Forms

[React Hook Form + Zod](#)

The same Zod schema validates in the browser and on the server.

Charts

[Recharts](#)

Composable, themeable, accessible — matches the token system.

Motion

[Framer Motion](#)

Used sparingly: state transitions in the capture flow, nothing decorative on the floor.

Realtime

[WebSocket](#) + [SSE fallback](#)

Live station status and analysis progress without polling.

Offline

[Service worker](#) + [IndexedDB](#)

Captures queue locally and replay on reconnect; the line never stops.

Testing

[Vitest](#) · [Testing Library](#) · [Playwright](#)

Unit and component coverage plus end-to-end runs of the full capture journey.

Accessibility

[WCAG 2.2 AA](#)

Keyboard-first station flow, 44px targets, verified contrast in both themes.

Performance

[LCP < 2.0s](#) · [INP < 200ms](#)

Budgeted and measured on the actual facility hardware, not a developer laptop.

07

[BUILD](#) [DETAIL](#)

The back end

API

[NestJS](#) · [TypeScript](#)

Modular and typed, same language as the front end, OpenAPI generated from the code.

Database

[PostgreSQL 16](#) + [Prisma](#)

Relational data with real constraints; row-level security gives hard tenant isolation.

Search

[Postgres GIN + trigram](#)

Fast serial and free-text search without running a second datastore on day one.

Object storage

[S3-compatible \(MinIO on prem\)](#)

Presigned uploads go straight from the station — images never transit the API.

Queue

[Redis + BullMQ](#)

Inference, PDF rendering, exports and webhooks all run as retryable jobs.

Inference service

[Python FastAPI + ONNX Runtime](#)

The ML ecosystem is Python; keeping it a separate service lets it scale on GPU alone.

Model serving

[TensorRT on the edge GPU](#)

Sub-second inference per image, on site, with no cloud round trip.

Auth

[OIDC / SAML + rotating JWT](#)

Enterprise SSO for staff, device-bound service tokens for stations.

Events

[Transactional outbox → webhooks](#)

Signed, retried, ordered delivery to the ERP — no lost events on a network blip.

Reports

[Headless Chromium worker](#)

Pixel-accurate branded PDFs from the same components the web app uses.

Observability

[OpenTelemetry](#) • [Grafana](#) • [Sentry](#)

Traces from station tap to model output; every slow inspection is explainable.

Infrastructure

[Docker](#) • [Terraform](#) • [GitHub Actions](#)

Reproducible environments, blue-green deploys, one-command facility rollout.

Backups

Nightly PITR · 30-day window

Images tier to cold storage after 90 days to keep running cost flat.

The core API is deliberately boring. All of the interesting engineering sits in the inference service and the grading engine — everything else should be predictable.

Images are the heavy part of this system: six full-resolution photographs per device, thousands of devices a day. They are uploaded straight to object storage with presigned URLs, so the API never becomes the bottleneck, and processing runs as queued jobs that can be retried, re-run against a newer model, or replayed after an outage.

Grading is a rule engine, not hard-coded logic. Defect weights, thresholds and per-client profiles live in the database and are versioned, which means a buyer can change their standard without a release — and any historic record can be re-evaluated against the profile that was live at the time.

DEPLOYMENT SHAPE

Control plane in the cloud, one edge node per facility. The edge node owns the cameras, the GPU and a local cache; it syncs records upward and pulls model updates downward. Adding a second site is a Terraform apply, not a project.

08

THE MODEL

How the detection engine is built and kept honest

We start from a pretrained segmentation backbone and fine-tune it on your own intake photography. The operator corrections captured in module 05 are the training signal — the system gets better because it is used, not because we rebuild it.

01

Capture

Six calibrated angles, fixed lighting, validated focus and exposure.

02

Pre-process

Device isolation, perspective correction, normalisation, tiling for small defects.

03

Inference

RT-DETR / YOLO-seg fine-tune, exported to ONNX, served on TensorRT.

04

Post-process

NMS, duplicate merging across angles, severity mapping, confidence gating.

05

Human review

Operator accepts, corrects, resizes or draws. Ground truth is created for free.

06

Retrain loop

Corrections queue into the training set; a candidate model must beat the incumbent on a frozen holdout before promotion.

WHAT WE NEED FROM YOU

- 1,500–3,000 representative intake photographs across your actual device mix
- A defect taxonomy sign-off session in week one — your names, your severities
- Two operators available for one labelling day per sprint
- Final camera rig and lighting spec by week three so training data matches production

HOW ACCURACY IS PROVEN

- A frozen holdout set is labelled once and never trained on
- Acceptance gate: recall ≥ 0.90 and precision ≥ 0.85 on the top ten defect classes

- Per-class scorecard published every sprint — no single blended number
- Anything below threshold routes to human review rather than being auto-graded

09

FOUNDATIONS

Core data model

Agreed and migrated in sprint one. Everything else is built on top of these.

Tenant

Client account; the isolation boundary for every query.

User · Role

Staff and buyer identities with the permission matrix.

Station · Camera

Physical rig, its six cameras and calibration profile.

Inspection

The central record: device, operator, timings, status, grades.

ImageAsset

One capture: angle, storage key, checksum, quality flags.

Detection

One defect: class, region, confidence, severity, review state.

DefectType

The taxonomy — editable, versioned, per tenant.

GradeProfile

Buyer-specific weights and thresholds used to grade.

GradeResult

Cosmetic, functional, health score, disposition, profile version.

AuditEvent

Append-only: actor, action, before, after, timestamp.

ModelVersion

Which model produced which result, with its scorecard.

Report · ShareLink

Generated certificates and their expiring public links.

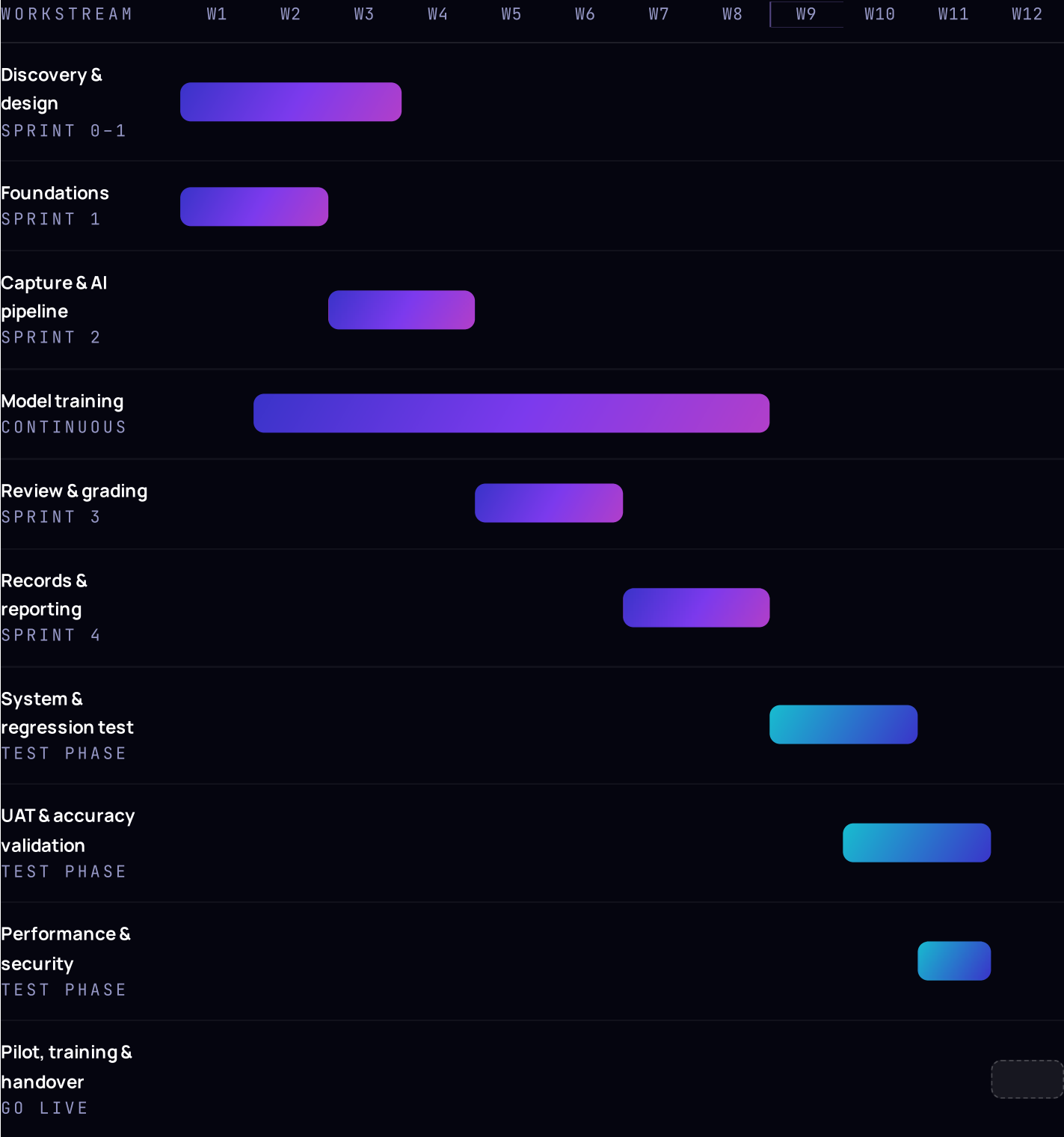
NON - NEGOTIABLES

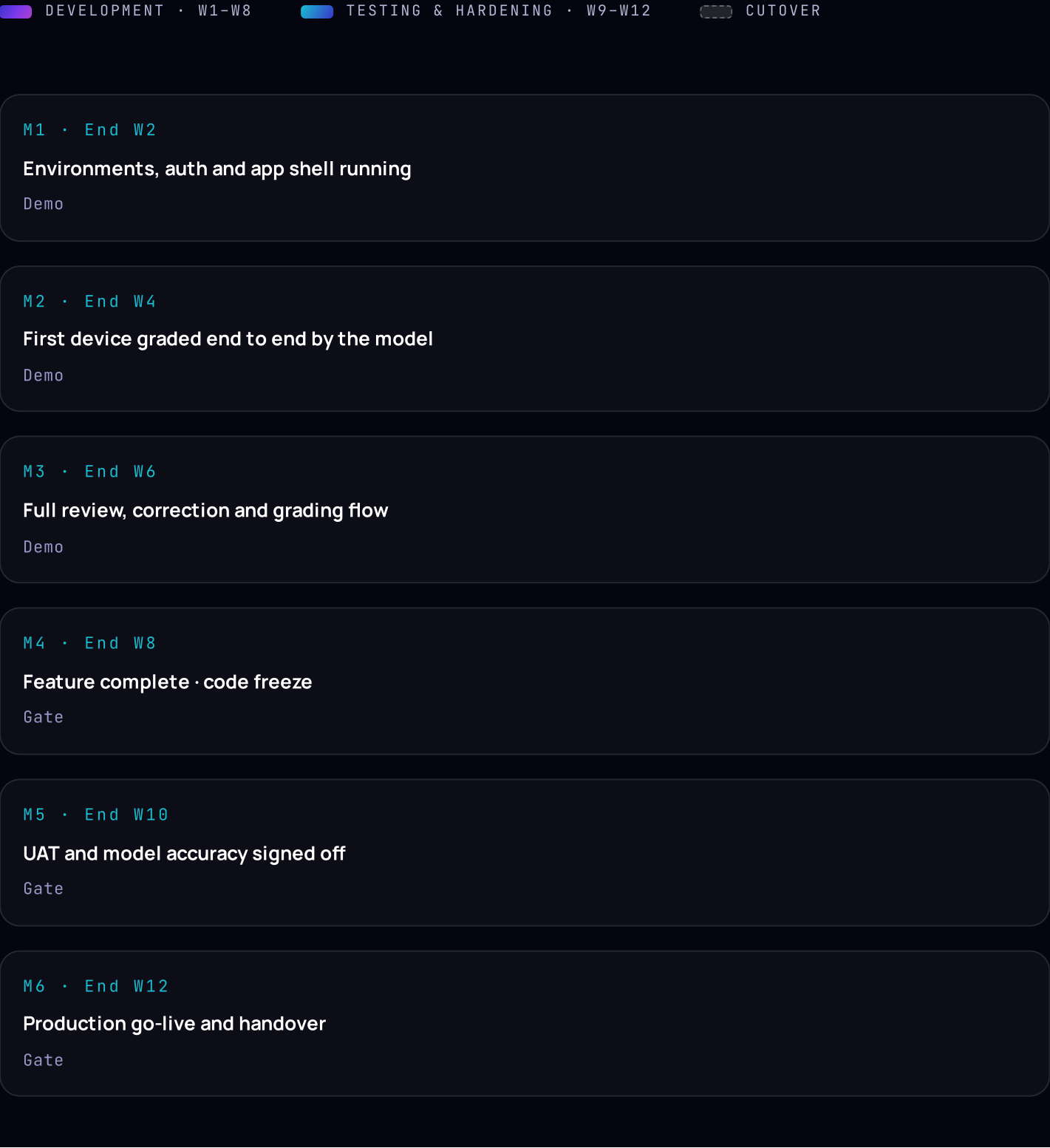
Security & compliance

- ☒ **Tenant isolation at the row**
Postgres RLS, not application-layer filtering — a bug in a query cannot leak another client's data.
-
- ☒ **Encryption in transit and at rest**
TLS 1.3 everywhere; AES-256 on volumes and object storage with documented key rotation.
-
- ☒ **Immutable audit trail**
Append-only table, exportable, covering every grade, override and defect edit.
-
- ☒ **Least privilege by default**
Permission matrix enforced server-side and covered by automated tests per role.
-
- ☒ **Retention & erasure**
Per-client retention windows, scheduled purge, and a right-to-erasure workflow.
-
- ☒ **Dependency & secret scanning**
SCA, SAST and secret detection run on every pull request; builds fail on new highs.
-
- ☒ **Third-party penetration test**
Booked for week eleven, with remediation time inside the plan rather than after it.
-
- ☒ **SOC 2-aligned controls**
Access reviews, change management and evidence export ready for an auditor.
-

Twelve weeks · eight to build, four to test

Four two-week sprints to feature complete, then a dedicated four-week hardening and acceptance phase. Testing is not a phase we hope to reach – it is scheduled, resourced and gated.





What lands, and when

Each sprint ends in a working demo on the staging environment and an exit criterion that is either met or not. There is no "90% done".

SPRINT 1 • WEEKS 1-2

BUILD

Foundations

- Discovery workshop, defect taxonomy sign-off, grading rules captured
- Design tokens and component library; station and back-office layouts
- Repository, CI/CD, staging and production environments, Terraform baseline
- Database schema and migrations; tenant isolation with row-level security
- Authentication, SSO wiring, role matrix and the permission test suite
- Application shell, navigation, station registry

EXIT CRITERIA

A user can sign in with SSO, land in the correct tenant, and see an empty but fully navigable application on staging. Labelling of the training set has begun.

SPRINT 2 • WEEKS 3-4

BUILD

Capture & the AI pipeline

- Station workflow: scan, verify, position, capture, analyse
- Camera controller integration, presigned uploads, quality validation and retake
- Inference service, job queue, and the first fine-tuned model in the loop
- Detection storage, confidence and severity mapping, model version stamping
- Offline capture queue and reconnect sync
- Live station status over WebSocket

EXIT CRITERIA

A real device is scanned, photographed from six angles and returned with model detections attached — the full happy path, on staging hardware.

Review, grading & records

- Annotated viewer: boxes, heat map, zoom-to-defect, cursor resize and draw tools
- Accept, reject, reclassify, merge, severity change — all audited
- Grading engine, health score, per-client grade profiles, disposition rules
- Override with reason codes and supervisor routing
- Inspection history grid with search, filters, saved views and bulk actions
- Automated regression suite starts running on every build

EXIT CRITERIA

An operator can correct the model's output and produce a defensible grade, with a complete audit trail behind every change.

Reporting, admin & integrations

- Inspection record tabs, image management, PDF certificate and share links
- CSV import wizard; CSV, XLSX, PDF and JSON exports
- Analytics dashboards and the scheduled digest
- Administration: clients, users, stations, taxonomy, grade profiles, branding
- Webhooks, ERP connector and scheduled buyer feeds
- Accessibility pass and performance budget enforcement

EXIT CRITERIA

All sixty features are implemented and demoed. **Code freeze.** Only defect fixes are merged from here.

System integration & regression

- Full SIT across station, API, inference and integrations
- Automated regression suite executed against a frozen build
- Data migration rehearsal and rollback drill
- Defect triage twice daily; severity 1 and 2 fixed same day

EXIT CRITERIA

Zero open severity-1 or severity-2 defects; regression suite green three consecutive runs.

UAT & model accuracy validation

- Structured UAT scripts run by your own operators and supervisors on real stock
- Model scored against the frozen holdout set; per-class scorecard published
- Grade agreement study – platform grade versus expert human grade
- Feedback log with a fix-or-defer decision recorded against every item

EXIT CRITERIA

UAT signed off by the sponsor; recall ≥ 0.90 and precision ≥ 0.85 on the top ten classes.

Performance, security & accessibility

- Load test at 3× expected peak throughput; soak test over 24 hours
- Third-party penetration test and remediation
- Failure drills: network loss, GPU offline, storage unavailable
- WCAG 2.2 AA audit of operator and supervisor journeys

EXIT CRITERIA

p95 API under 300ms at peak; no critical or high security findings open; accessibility audit passed.

Pilot, training & handover

- Pilot run on one live line, monitored, with the old process running alongside
- Final defect burn-down and release candidate sign-off
- Operator, supervisor and admin training plus recorded walkthroughs
- Runbooks, architecture docs, API reference and model retraining guide
- Production cutover and thirty days of hypercare

EXIT CRITERIA

Pilot line running on the platform for five consecutive days without a severity-1 incident. Handover accepted.

12

HONEST ASSESSMENT

Risks and how we hold them

RISK	LEVEL	MITIGATION
Not enough labelled training data	HIGH	Start from a pretrained backbone; run a labelling sprint from week one; use active learning so operator corrections in weeks 5–10 close the gap. Anything under the confidence threshold routes to a human rather than guessing.
Camera rig or GPU hardware arrives late	HIGH	A mock camera driver and a recorded image set are built in sprint 2, so software delivery is never blocked by hardware. Hardware spec is frozen at week three.
Lighting or positioning varies between sites	MEDIUM	Fixed rig geometry, a per-station calibration profile, and automatic exposure and focus validation that refuses a bad frame before it reaches the model.
ERP integration access is slow to arrange	MEDIUM	Contract-first: the connector is built against an agreed schema with a stub, and swapped to the live endpoint when credentials arrive. Not on the critical path.

RISK	LEVEL	MITIGATION
Grading rules change during the build	MEDIUM	Grading is configuration, not code. Rules live in versioned profiles that can be edited without a release and replayed against historic records.
Scope grows inside a fixed twelve weeks	MEDIUM	The sixty features above are the contract. New requests are logged, estimated and scheduled after go-live, or swapped in against something of equal size.
Operator adoption on the floor	LOW	Two of your operators are involved from week one, sit in every demo, and run UAT themselves. The people who will use it help shape it.

ASSUMPTIONS

- You provide sample devices covering your real intake mix, plus 1,500–3,000 photographs for training
- Camera rig, lighting and GPU hardware are procured by you and on site by week three
- Two operators and one supervisor are available for one day per sprint
- One integration endpoint (ERP or WMS) is in scope; further systems are change requests
- Sponsor attends a weekly one-hour review and turns decisions around within two working days
- Cloud hosting and third-party licences are billed to you at cost

OUT OF SCOPE FOR V1

- Native mobile applications – the web app is responsive and installable
- Automated repair-cost estimation and parts pricing
- Marketplace listing sync (eBay, Back Market and similar)
- Languages other than English in the interface
- Robotic device handling and conveyor control
- Functional testing of the device itself – battery cycles, port continuity, screen response

Getting to week one

Four things stand between this document and a kick-off. None of them take more than a week.

01

Scope sign-off

Confirm the sixty features, the assumptions and what is out of scope for v1.

02

Commercials & contract

Statement of work, payment schedule tied to the six milestones, and signature.

03

Access & hardware

Environments, SSO tenant, ERP contact, and a purchase order for the camera rig.

04

Discovery workshop

One day on site: taxonomy, grading rules, and photographs of your real stock.

ACCEPTED FOR THE CLIENT

Hugo Martinez

Project sponsor · Date

PROPOSED BY

